DHANALAKSHMI SRINIVASAN UNIVERSITY

Samayapuram, Tiruchirappalli, Tamil Nadu – 621 112

SCHOOL OF COMPUTER APPLICATIONS • DEPARTMENT OF MCA

CONTINUOUS ASSESSMENT TEST - 1 (ODD SEMESTER)

Register No.		**Year / Sem**	I / I MCA
Course Title	**DATA STRUCTURES**	**Course Code**	26MCA103
Time Allowed	90 Minutes (1.5 Hours)	**Max. Marks**	**50 Marks**
Course Outcomes	**CO1:** Understand fundamental data structures, arrays, and linked lists with operations. **CO2:** Apply stack and queue principles in evaluation, algorithms, and applications.		

PART – A : CONCEPTUAL QUESTIONS (10 × 2 = 20 MARKS)

Answer ALL questions

Q.No	Question Description (From Question Bank)	BT Level	CO	Marks
1	Define a data structure. Give two examples.	BT1 (Remember)	CO1	**2**
2	What is an Abstract Data Type (ADT)? Give one example.	BT1 (Remember)	CO1	**2**
3	Define time complexity and space complexity.	BT1 (Remember)	CO1	**2**
4	Differentiate between Big-O and Big-Ω notations.	BT2 (Understand)	CO1	**2**
5	What is a doubly linked list? How does it differ from a singly linked list?	BT2 (Understand)	CO1	**2**
6	Define a stack.	BT1 (Remember)	CO2	**2**
7	What are stack overflow and stack underflow conditions?	BT1 (Remember)	CO2	**2**
8	Convert the infix expression `A + B * C` into postfix notation.	BT2 (Understand)	CO2	**2**
9	What is the FIFO principle?	BT1 (Remember)	CO2	**2**
10	State one advantage of a circular queue over a linear queue.	BT2 (Understand)	CO2	**2**

PART – B : APPLICATION QUESTIONS (2 × 15 = 30 MARKS)

Answer ALL questions (Either / Or Choice)

Q.No	Question Description (From Question Bank)	BT Level	CO	Marks
11.a	Explain singly linked lists, including node representation and algorithms for creation, traversal, insertion, deletion, searching, and updating operations.	BT3 (Apply)	CO1	**15**
	[OR]			
11.b	Compare arrays and linked lists based on memory representation, access, insertion, deletion, searching, advantages, limitations, and applications.	BT4 (Analyze)	CO1	**15**
12.a	Explain the procedure for converting an infix expression into postfix using a stack with a suitable example.	BT3 (Apply)	CO2	**15**
	[OR]			

Q.No	Question Description (From Question Bank)	BT Level	CO	Marks
12.b	Explain the implementation of a circular queue and its enqueue and dequeue operations with algorithms.	BT3 (Apply)	CO2	**15**

★ Key Examination Preparation Checklist:

- **Unit I (Part-B Alternatives):** Also revise *Asymptotic Notations (Big-O, Big-Ω, Big-Θ)* and *Insertion/Deletion algorithms in Singly/Doubly Linked List*.
- **Unit II (Part-B Alternatives):** Also revise *Postfix Expression Evaluation using Stack* and *Linear Queue vs Circular Queue memory comparison*.
- **Representation Tip:** Draw clear pointer/link diagrams for linked lists, tabular tracking for stack evaluation, and circular array diagrams for queues to secure full marks.